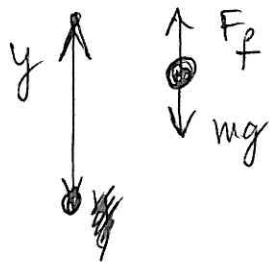
$\boxed{1}$ $ma = F_g + F_f$



$$m\frac{dv}{dt} = -mg - \gamma v$$

$$dv = \left(-g - \frac{\gamma}{m}v\right)dt$$

$$\frac{dv}{g + \frac{\gamma}{m}v} = -dt$$

$$\frac{m}{\gamma}\, d\ \ln\left(g + \frac{\gamma}{m}v\right) = -dt$$

$$\ln\frac{g + \frac{\gamma}{m}v(t)}{g + \frac{\gamma}{m}v(0)} = -\frac{\gamma}{m}t$$

$$g + \frac{\gamma}{m}v(t) = \left(g + \frac{\gamma}{m}v(0)\right)e^{-\frac{\gamma t}{m}}$$

$$v(t) = -\frac{mg}{\gamma} + \left(v(0) + \frac{mg}{\gamma}\right)e^{-\frac{\gamma t}{m}}$$

$$y(t) - y(0) = \int_0^t v(t)\,dt = -\frac{mg}{\gamma}t + \left(v_0 + \frac{mg}{\gamma}\right)\int_0^t e^{-\frac{\gamma t}{m}}dt$$

$$\int_0^t e^{-\frac{\gamma t}{m}}dt = -\frac{m}{\gamma}\left(e^{-\frac{\gamma t}{m}} - 1\right)$$

Thus $y(t) = y(0) - \frac{mgt}{\gamma} + \frac{m}{\gamma}\left(v_0 + \frac{mg}{\gamma}\right)\left(1 - e^{-\frac{\gamma t}{m}}\right)$

So that at long times, $\frac{t\gamma}{m} \gg 1$,

$$y(t) = y_0 - \frac{mgt}{\gamma}$$

particle is moving with constant velocity $-\frac{mg}{\gamma}$.

$\boxed{2}$ Minimize $f(x,y,z) = x^2+2y^2+3z^2+2xy+2xz$ subject to condition that

$$g(x,y,z) = x^2+y^2+z^2-1 = 0.$$

Since the points in question belong to the surface of the unit sphere, this is the same as minimizing function $\tilde{f}(x,y,z) = 1+y^2+2z^2+2x(y+z)$

Construct $\varphi = y^2+2z^2+2x(y+z) - \lambda(x^2+y^2+z^2-1)$

$$\begin{cases} \varphi'_x = 0 = 2(y+z) - 2\lambda x & (1) \\ \varphi'_y = 0 = 2y+2x-2\lambda y \Rightarrow y = \dfrac{-x}{1-\lambda} \\ \varphi'_z = 0 = 4z+2x-2\lambda z \Rightarrow z = \dfrac{-x}{2-\lambda} \\ \varphi'_\lambda = 0 = x^2+y^2+z^2-1 & (2) \end{cases}$$

(1): $y+z-\lambda x = 0$

$$-\frac{x}{1-\lambda} - \frac{x}{2-\lambda} - \lambda x = 0$$

$$\frac{1}{1-\lambda} + \frac{1}{2-\lambda} + \lambda = 0$$

($x=0$ implies $y=z=0$ which does not satisfy constraint (2))

$$\lambda(1-\lambda)(2-\lambda) + 1-\lambda + 2-\lambda = 0$$

$$\lambda(2-3\lambda+\lambda^2) + 3-2\lambda = 0$$

$$\lambda^3 - 3\lambda^2 + 3 = 0$$

Roots are: $\lambda_1 = -0.879385$, $\lambda_2 = 1.3473$, $\lambda_3 = 2.53209$

Corresponding x is found from (2):

$$x^2\left[1+\frac{1}{(1-\lambda)^2}+\frac{1}{(2-\lambda)^2}\right]=1$$

$$x=\pm\frac{|1-\lambda|\,|2-\lambda|}{\sqrt{(1-\lambda)^2+(2-\lambda)^2+(1-\lambda)^2(2-\lambda)^2}}$$

Then $f(x,y,z)=1+y^2+2z^2+2x(y+z)$

From (1) $y+z=\lambda x$, thus

$$f=1+\left(\frac{x}{1-\lambda}\right)^2+2\left(\frac{x}{2-\lambda}\right)^2+2x\cdot\lambda x=$$

$$=1+x^2\left\{2\lambda+\frac{2}{(2-\lambda)^2}+\frac{1}{(1-\lambda)^2}\right\}=$$

$$=1+\frac{(1-\lambda)^2(2-\lambda)^2}{(1-\lambda)^2+(2-\lambda)^2+(1-\lambda)^2(2-\lambda)^2}\left[2\lambda(2-\lambda)^2(1-\lambda)^2+2(1-\lambda)^2+\right.$$

$$\left.+(2-\lambda)^2\right]\frac{1}{(2-\lambda)^2(1-\lambda)^2}=$$

$$=1+\frac{2\lambda(2-\lambda)^2(1-\lambda)^2+2(1-\lambda)^2+(2-\lambda)^2}{(1-\lambda)^2+(2-\lambda)^2+(1-\lambda)^2(2-\lambda)^2}$$

We find that minimum happens at $\lambda_1=-0.879385$ when $f(x,y,z)=0.120615$.

$\boxed{3}$ $$S = \int ds = \int_{x_1}^{x_2} \sqrt{1+(y')^2}\, dx$$

$$\frac{\partial L}{\partial y} = 0 \Rightarrow \frac{d}{dx}\frac{\partial L}{\partial y'} = 0$$

$\Rightarrow$ ~~const~~ integral of motion $\frac{\partial L}{\partial y'} = \frac{y'}{\sqrt{1+(y')^2}} = C$

$$(y')^2 = C^2\left(1+(y')^2\right)$$

$$(y')^2 = \frac{C^2}{1-C^2}$$

$$y' = \frac{C}{\sqrt{1-C^2}} = \alpha$$

$$dy = \alpha\, dx \Rightarrow y = \alpha x + \beta$$

Boundary conditions: $\left.\begin{array}{l} y(x_1) = y_1 \\ y(x_2) = y_2 \end{array}\right\}$ $\begin{array}{l} \alpha x_1 + \beta = y_1 \\ \alpha x_2 + \beta = y_2 \end{array}$

$$\alpha(x_2 - x_1) = y_2 - y_1, \quad \alpha = \frac{y_2 - y_1}{x_2 - x_1}$$

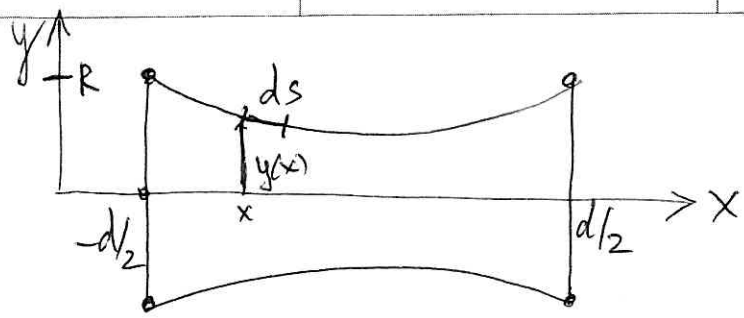
$$\beta = y_1 - \alpha x_1 = y_1 - \frac{y_2 - y_1}{x_2 - x_1} x_1 =$$

$$= \frac{y_1(x_2 - x_1) - (y_2 - y_1)x_1}{x_2 - x_1} = \frac{y_1 x_2 - y_2 x_1}{x_2 - x_1}$$

Straight line eq: $$y(x) = \frac{y_2 - y_1}{x_2 - x_1} x + \frac{y_1 x_2 - y_2 x_1}{x_2 - x_1}$$

$\boxed{4}$

(a)



Area of segment ds : $2\pi\, y(x)\, ds$

Minimize $A[y(x)] = \int_{x_1}^{x_2} y(x)\sqrt{1+(y')^2}\,dx = \int_{-d/2}^{d/2} L(y,y')\,dx$

subject to BC : $y(-d/2) = y(d/2) = R$

Since L does not depend on x there is conserved quantity $H = y'\dfrac{\partial L}{\partial y'} - L$

$$H = y'\frac{y'y}{\sqrt{1+(y')^2}} - y\sqrt{1+(y')^2} = \frac{y}{\sqrt{1+(y')^2}}\left\{(y')^2 - (1+(y')^2)\right\}$$

$$= \frac{-y}{\sqrt{1+(y')^2}}$$

Hence $\dfrac{1+(y')^2}{y^2} = \dfrac{1}{H^2}$

$$(y')^2 = \frac{y^2}{H^2} - 1 = \frac{y^2-H^2}{H^2}$$

$$y' = \frac{dy}{dx} = \pm\frac{\sqrt{y^2-H^2}}{H}$$

The sign choice should not matter as by symmetry we expect $y'<0$ near $-d/2$ and $y'>0$ near $\frac{d}{2}$.

Choose (+) for concreteness:

$$\frac{H\,dy}{\sqrt{y^2-H^2}} = dx$$

let $y = H\cosh(u)$: $y^2-H^2 = H^2\left(\cosh^2(u)-1\right) = H^2\sinh^2(u)$

$$\int \frac{H^2\, d(\cosh(u))}{H\sinh(u)} = x-x_0$$

But $d(\cosh(u)) = \sinh(u)$

$$\Rightarrow \quad u = \frac{x-x_0}{H}$$

$$\Rightarrow \quad y = H\cosh\left(\frac{x-x_0}{H}\right)$$

BC: $\left.\begin{aligned} y(+d/2) &= H\cosh\left(\frac{x_0-d/2}{H}\right) = R \\ y(-d/2) &= H\cosh\left(\frac{x_0+d/2}{H}\right) = R \end{aligned}\right\}$ must be $x_0 = 0$.

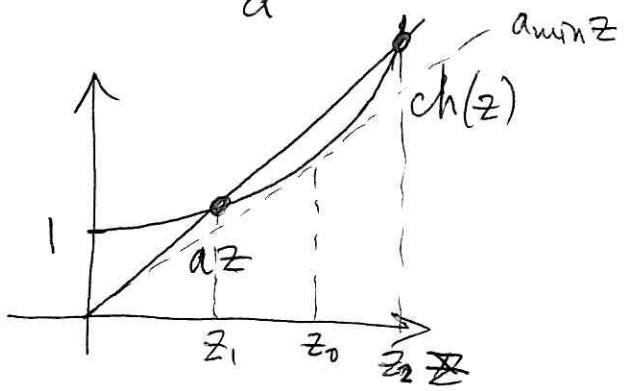
$$\Rightarrow \quad y(x) = H\cosh\left(\frac{x}{H}\right)$$

(b) Again BC: $\frac{R}{H} = \cosh\left(\frac{d}{2H}\right)$

let $\frac{d}{2H} = z \Rightarrow \frac{R}{H} = \frac{R}{(d/2z)} = \frac{2R}{d} z$

Thus we need to analyze

$$\frac{2R}{d} z = \cosh(z) \; ; \quad a \equiv \frac{2R}{d}$$



For $a > a_{min}$ there are <u>two</u> crossings.

The smallest a_{min} corresponds to straight line touching $\cosh(z)$ at a single point: when this happens the two curves have the same slope $\Rightarrow a_{min} z_0 = ch(z_0)$

$$a_{min} = sh(z_0)$$

$$\Rightarrow z_0 = \frac{ch(z_0)}{sh(z_0)} = \coth(z_0).$$

This is solved by

$$z_0 = 1.19968 \Rightarrow a_{min} = sh(z_0) = 1.50888$$

$$\Rightarrow \left(\frac{2R}{d}\right)_{min} \cong 1.509 \Rightarrow x_{min} = 0.7545 \qquad x = \frac{R}{d}$$

$\Rightarrow$ the distance between the rings $\frac{d}{R} \leq 1.325$.

For $\frac{d}{R} < 1.325$ solution exists.

It must be that out pair of z's solving

$$az = ch(z)$$

one corresponds to the <u>minimal</u> surface

and another to the <u>maximal</u> surface.

Back to area: $A = \int_{-d/2}^{d/2} y\sqrt{1+(y')^2}\,dx =$

$$= \int_{-d/2}^{d/2} H\, ch\left(\frac{x}{H}\right)\sqrt{1+sh^2\left(\frac{x}{H}\right)}\,dx = (x = yH) =$$

$$= H^2 \int_{-\frac{d}{2H}}^{d/2H} ch(y)\, ch(y)\,dy = H^2 \int_{-z}^{z} ch^2(y)\,dy$$

where $z = \frac{d}{2H} \Rightarrow H = \frac{d}{2z}$

$$\Rightarrow A = H^2\left(z + \cosh(z)\sinh(z)\right) =$$

$$= \left(\frac{d}{2z}\right)^2\left(z + \frac{2Rz}{d}\sqrt{\left(\frac{2Rz}{d}\right)^2 - 1}\right) = \frac{d^2}{4}\,\frac{1+\frac{2R}{d}\sqrt{\left(\frac{2Rz}{d}\right)^2-1}}{z}$$

is decreasing function of $z \Rightarrow$ we get least area from larger of two roots: z_2.